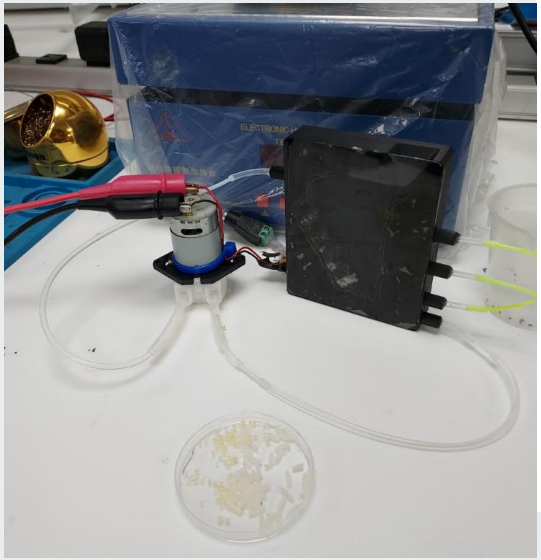
Super Filter

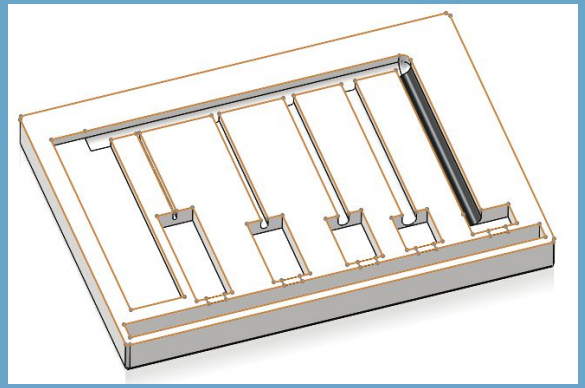
AI Vision Tracking for Entomological Library Creation with Solid & Liquid Flow Chips

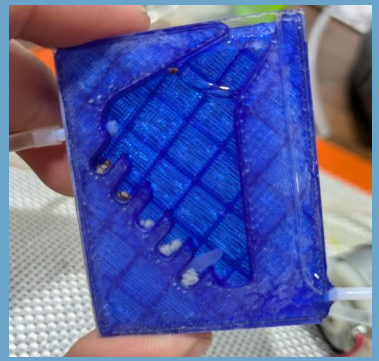




Members:
Renaud Renault
Dale Herzog
Mohamed

Outline





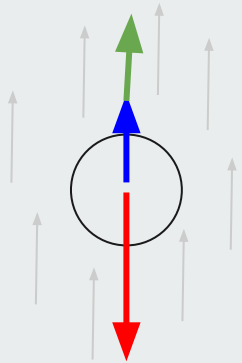
The Problem: Traditional insect sorting techniques using filters have a hard time managing entrained samples and surface tension forces and debris.

Solution: Use the troublesome physical forces to our advantage in separate liquid and solid sorting chips.

Option 1 : Sorting by size using sedimentation.

Principle :

- Forces : (gravity - archimede) downwards, drag force in the direction of the fluid flow direction (opposite to the object).

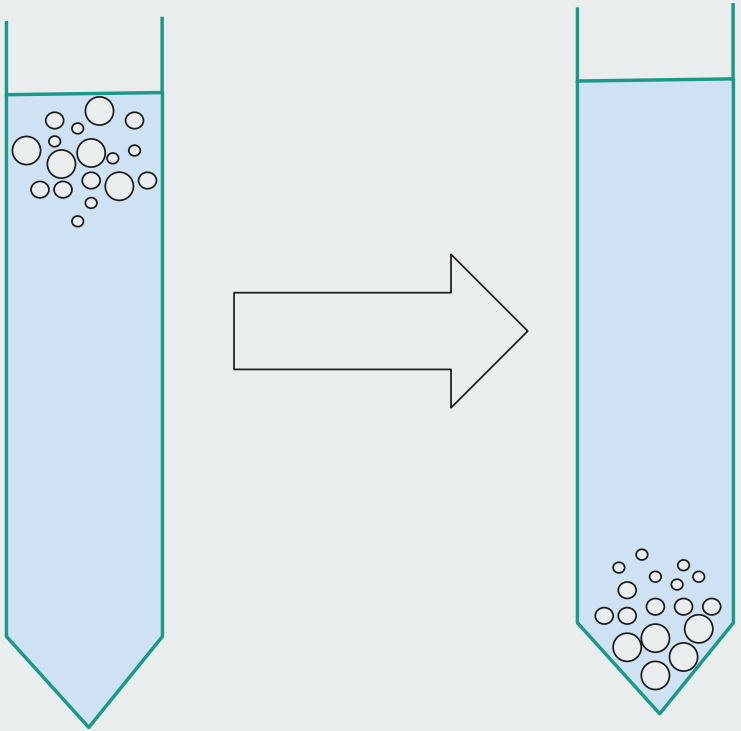


Option 1 : Sorting by size using sedimentation.

- The drag force is proportional to cross section (area).
- The other forces depends on mass (volume)

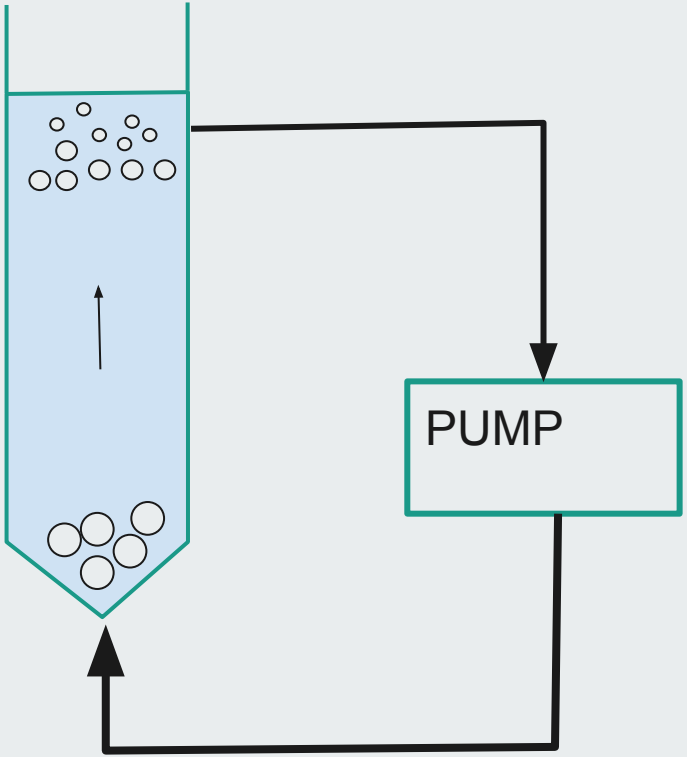
Smaller particles (at fixed density) sediment at a slower speed because their S/V is greater (drag dominates).

Sedimentation works for sorting

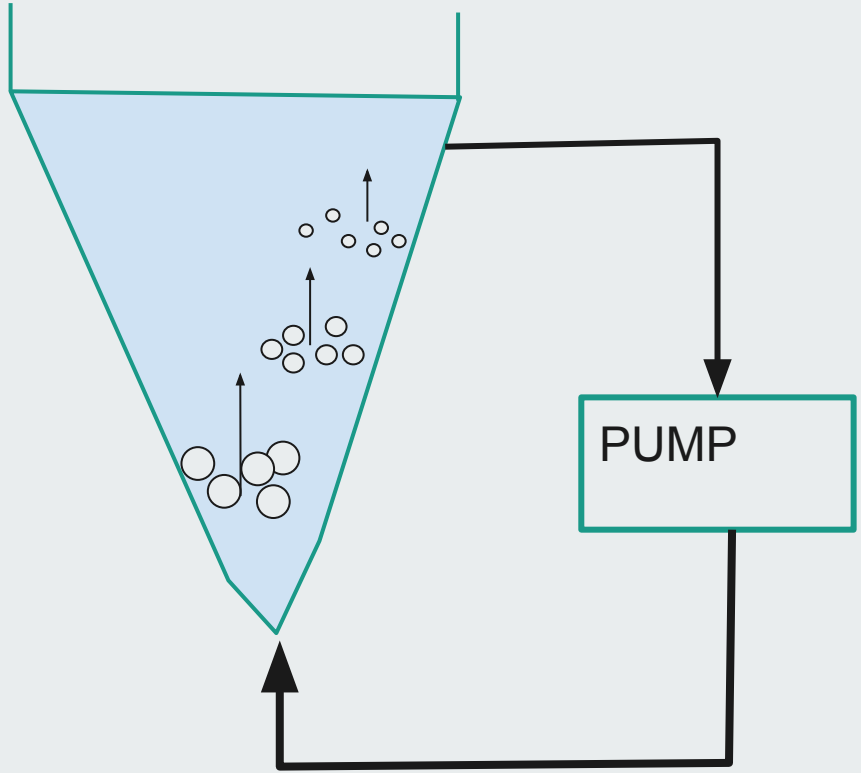


But how do we retrieve the different sizes ?

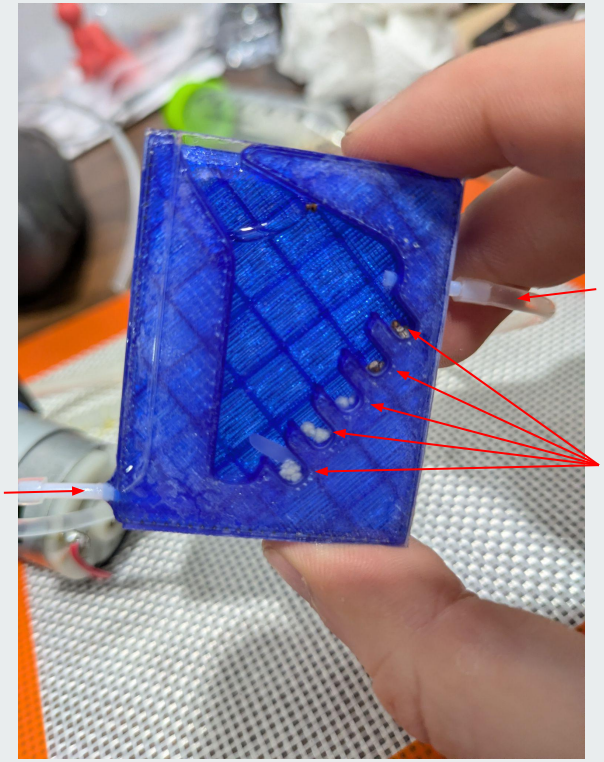
Adding flow only separates the mix but only into two groups



Adding varying section size allows continuous separation



Results for the Proof of concept



NB: Collection chambers are added to retrieve the samples

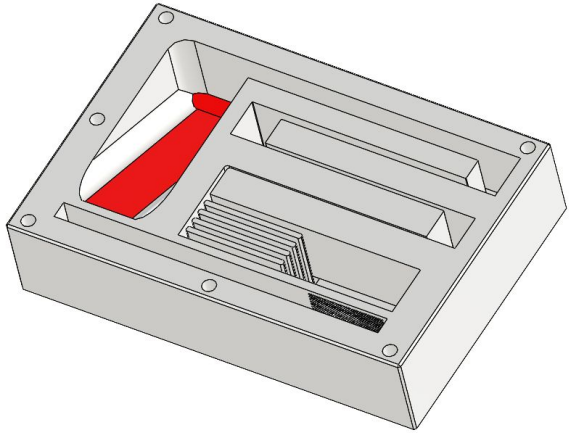
Liquid Sorting

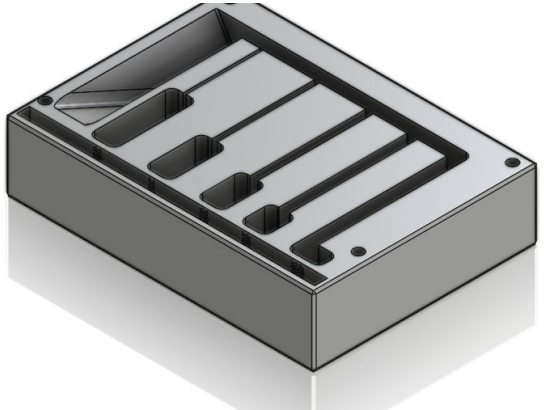
1. Separating bugs in liquid is basically impossible with metal tools.
2. Bugs are damaged
3. Different size bugs stick to each other

Key Concepts for the Fluidic Chip

Use fluid properties to our advantage.

1. Turbulence
2. The distributed force application of fluids
3. Reducing Surface Tension
4. “Coin Machine” Flow Path Sorting

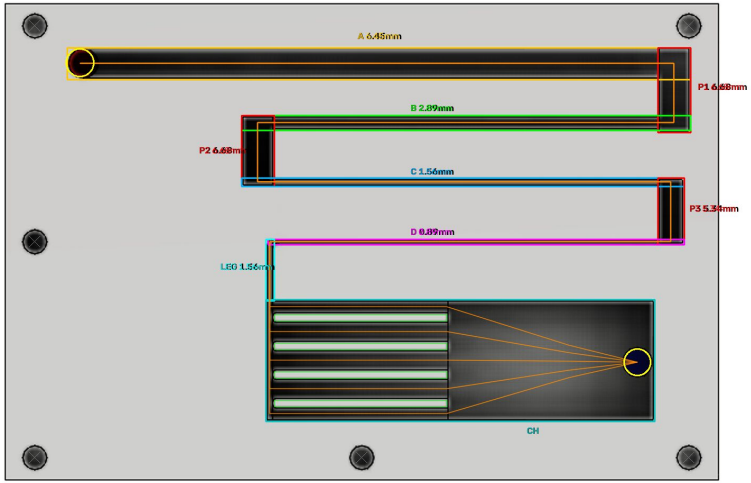




CAD to Test





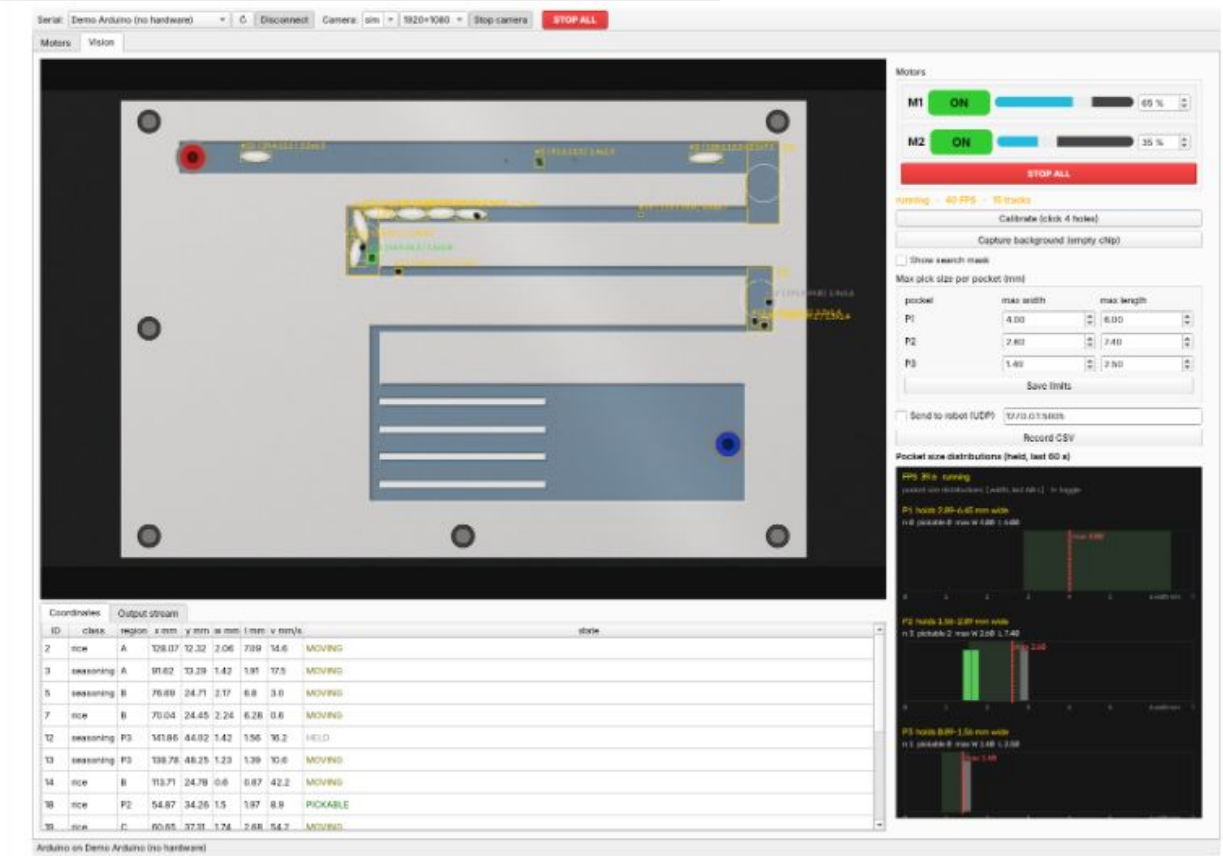


traced vector (color) over CAD top view | scale assumes plate 150 x 100 mm | 1 px = 0.22 mm

Optical inspection

As part of an automated sorting system, target items (insects/bugs) fee them through an optical inspection zone inside a clear tube. A high-speed camera captures a single-item snapshot per transit. The computer vision subsystem accomplish a real-time classify the item instance into its target group.

Test to Point Collection



Questions?